

Taiyu Zhang

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SUMMARY

- **Education** — PhD candidate at KU Leuven Computer Science; M.S. from the Department of Electronic Engineering, Tsinghua University; B.S. from USTC's School of Information.
- **Research areas** — Human-Computer Interaction; generative foundation models.
- **Project experience** — AI Agents; XR interaction; Language model generation; Full-body motion generation; World models.
- **Interests** — Video games; finding good restaurants.

EXPERIENCE

- Research Intern, Multimodal Interaction — ByteDance PICO** Jul 2026 – Present
- Research on the synthetic-data pipeline for hand-gesture tracking on XR devices.
- Research Intern — Moxin Technology** Mar 2026 – May 2026
- Autoregressive training and acceleration of long-video generation models.
- NLP Researcher — iFlyTek** 2021 – 2022
- Multiple national-level R&D projects; main owner of the multilingual semantic-understanding component of the iFlyTek Digital Human — data curation, model training, and evaluation.
 - Deployed in the 2022 Beijing Winter Olympics and other state-level events.
 - Long-sentence-completion GPT pretraining for the iFlyTek input method.

SELECTED PROJECTS

- Hand-Gesture Data Pipeline — Diffusion data generation + multimodal VLM annotation** 2026.07 – Present
- Diffusion-based generation of hand-gesture video data, widening training coverage past what captured sessions reach: unusual hand shapes, lighting, skin tones, viewing angles, and occlusion.
 - Annotation of the gesture video data with the QwenVL multimodal understanding model, together with review of annotation quality.
- LingBot Distillation — Lead researcher, 14B video diffusion → 4-step student on Ascend** 2026.03 – 2026.05
- Reproduced LingBot's distillation on Huawei Ascend (CANN); teacher: camera-trajectory control + 1000-frame long-video coverage. 40–50-step teacher → 4-step student; > 50-min generation → < 4 min.
 - Resolved self-forcing memory bottleneck with two-pass forward + activation recompute (gradient-accumulation style); proposed a new ODE-initialization that uses training data to init student directly from teacher, removing explicit ODE-pair generation and stabilising distillation.
- Claw-AI-Lab — Co-developer** 2025 – Present
- Multi-agent autonomous research platform: one prompt → paper, code, figures, and experiment logs. 1.3k+ GitHub stars.
 - Contributed across the multi-agent scheduler, the local-execution harness, and the real-time monitoring dashboard.
- iFlyTek IME Long-Sentence Completion — Lead researcher, GPT pretraining from scratch** 2021 – 2022
- Built a 400 GB → 40 GB pretraining-data pipeline: deduplication, language and quality filtering, topic balancing, formatting normalization.
 - Pretrained a GPT-style language model from scratch on the cleaned 40 GB Chinese corpus.
 - Shipped in the iFlyTek input method, powering long-text completion and poetry-creation features in production.
- PhD Research: LLM Agents in Immersive Environments — First author on all three** 2023 – Present
- **Focus Agent** (IVA 2024) — LLM-driven virtual focus group: each participant is an agent with persona, reasoning style, and memory; a moderator steers the session toward an interview goal.
 - **AI Game NPC** (SUI 2026) — LLM companion in a Unity 6 VR puzzle game on Apple Vision Pro: Whisper perception, GPT-4 cognition querying game-engine state directly, TTS-plus-action executor. A 24-participant study across four assistance levels found assistance reduces workload but full autonomy hurts experience, with four player types diverging sharply.
 - **Coembodiment AI** — first validation of users' subjective experience when a human and an AI share control of one virtual body; a non-linear adaptive control function outperforms gate / linear baselines, generalising to human-robot collaborative control.

EDUCATION

- Ph.D., Computer Science — KU Leuven** 2022 – Present
- Research direction: 3D multimodal intelligent agents in immersive environments. Advisor: Prof. Adalberto L. Simeone.
- M.S., Electronic Engineering — Tsinghua University** 2018 – 2021
- Thesis: ML-based quantitative analysis of soil elements via LIBS. Advisor: Prof. Xiaohong Ma.
- B.S., Electronic Engineering & Information Science — USTC** 2014 – 2018
- Thesis: SRCNN-based image super-resolution (Outstanding Undergraduate Thesis). Advisor: Prof. Jian Tang.

SELECTED PUBLICATIONS

- “What Can I Do for You”: How Should AI Companions Provide Assistance to Players in Virtual Reality Games.
Zhang, T., et al. *ACM Symposium on Spatial User Interaction (SUI)*, 2026.
- MoWorld: A Flash World Model.**
Team Moxin, incl. Zhang, T. Technical report, [arXiv:2607.06216](https://arxiv.org/abs/2607.06216), 2026.
- Focus Agent: LLM-Powered Virtual Focus Group.**
Zhang, T., et al. *Proceedings of the 24th ACM International Conference on Intelligent Virtual Agents (IVA)*, 2024. [arXiv:2409.01907](https://arxiv.org/abs/2409.01907).